



Mersenne Hierarchical Decomposition

See how p is built from smaller Mersenne exponents (hex labels)

21701, 44497, 756839

p = 3 (base 3) · Label: 1

✓ KNOWN MERSENNE

0

pattern bits



p is prime

Hierarchical Decomposition:

1 → p = 3 (pattern: 1)

↳ Built from:

↻ Cycle: 1

p = 5 (base 2) · Label: 2

✓ KNOWN MERSENNE

1

pattern bits



p is prime

Hierarchical Decomposition:

2 → p = 5 (pattern: 1)

↳ Built from:

↻ Cycle: 2

p = 7 (base 3) · Label: 3

✓ KNOWN MERSENNE

1

pattern bits



p is prime

Hierarchical Decomposition:

3 → p = 7 (pattern: 1)

↳ Built from:

↻ Cycle: 3

p = 13 (base 3) · Label: 5

✓ KNOWN MERSENNE

2

pattern bits

✓

p is prime

Hierarchical Decomposition:

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

p = 17 (base 2) · Label: 6

✓ KNOWN MERSENNE

3

pattern bits

✓

p is prime

Hierarchical Decomposition:

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

p = 19 (base 2) · Label: 7

✓ KNOWN MERSENNE

3

pattern bits

✓

p is prime

Hierarchical Decomposition:

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

p = 31 (base 3) · Label: 8

✓ KNOWN MERSENNE

3

pattern bits



p is prime

Hierarchical Decomposition:

8 → p = 31 (pattern: 111)

↳ Built from:

↻ Cycle: 8

p = 61 (base 3) · Label: 9

✓ KNOWN MERSENNE

4

pattern bits



p is prime

Hierarchical Decomposition:

9 → p = 61 (pattern: 1101)

↳ Built from:

↻ Cycle: 9

p = 89 (base 2) · Label: a

✓ KNOWN MERSENNE

5

pattern bits



p is prime

Hierarchical Decomposition:

a → p = 89 (pattern: 11001)

↳ Built from:

↻ Cycle: a

p = 107 (base 3) · Label: b

✓ KNOWN MERSENNE

5

pattern bits



p is prime

Hierarchical Decomposition:

b → p = 107 (pattern: 01011)

↳ Built from:

↻ Cycle: b

p = 127 (base 3) · Label: c

KNOWN MERSENNE

5

pattern bits



p is prime

Hierarchical Decomposition:

c → p = 127 (pattern: 11111)

↳ Built from:

↻ Cycle: c

p = 521 (base 2) · Label: 0066

KNOWN MERSENNE

8

pattern bits



p is prime

Hierarchical Decomposition:

0066 → p = 521 (pattern: 00001001)

↳ Built from:

Filler zero

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

p = 607 (base 2) · Label: 578

✓ KNOWN MERSENNE

8

pattern bits

✓

p is prime

Hierarchical Decomposition:

578 → p = 607 (pattern: 01011111)

↳ Built from:

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

8 → p = 31 (pattern: 111)

↳ Built from:

↻ Cycle: 8

p = 1279 (base 2) · Label: 788

✓ KNOWN MERSENNE

9

pattern bits

✓

p is prime

Hierarchical Decomposition:

788 → p = 1279 (pattern: 01111111)

↳ Built from:

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↻ Cycle: 8

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↻ Cycle: 8

p = 2203 (base 2) · Label: 6077

✓ KNOWN MERSENNE

10

pattern bits

✓

p is prime

Hierarchical Decomposition:

$6077 \rightarrow p = 2203$ (pattern: 0010011011)

↳ Built from:

$6 \rightarrow p = 17$ (pattern: 001)

↳ Built from:

↻ Cycle: 6

Filler zero

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↻ Cycle: 7

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↻ Cycle: 7

p = 2281 (base 2) · Label: 696

✓ KNOWN MERSENNE

10

pattern bits

✓

p is prime

Hierarchical Decomposition:

696 → p = 2281 (pattern: 0011101001)

↳ Built from:

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

9 → p = 61 (pattern: 1101)

↳ Built from:

↺ Cycle: 9

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 3217 (base 3) · Label: 6606

✓ KNOWN MERSENNE

10

pattern bits

✓

p is prime

Hierarchical Decomposition:

6606 → p = 3217 (pattern: 0010010001)

↳ Built from:

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 4253 (base 2) · Label: 0669

✓ KNOWN MERSENNE

11

pattern bits



p is prime

Hierarchical Decomposition:

0669 → p = 4253 (pattern: 00010011101)

↳ Built from:

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

9 → p = 61 (pattern: 1101)

↳ Built from:

↺ Cycle: 9

p = 4423 (base 2) · Label: 650008

KNOWN MERSENNE

11

pattern bits



p is prime

Hierarchical Decomposition:

650008 → p = 4423 (pattern: 00101000111)

↳ Built from:

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

Filler zero

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

p = 9689 (base 2) · Label: 50876

✓ KNOWN MERSENNE

12

pattern bits

✓

p is prime

Hierarchical Decomposition:

$50876 \rightarrow p = 9689$ (pattern: 010111011001)

↳ Built from:

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

$6 \rightarrow p = 17$ (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 9941 (base 2) · Label: 77555

✓ KNOWN MERSENNE

12

pattern bits

✓

p is prime

Hierarchical Decomposition:

77555 → p = 9941 (pattern: 011011010101)

↳ Built from:

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

p = 11213 (base 2) · Label: 158075

✓ KNOWN MERSENNE

12

pattern bits

✓

p is prime

Hierarchical Decomposition:

158075 → p = 11213 (pattern: 101111001101)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

↻ Cycle: 1

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

p = 19937 (base 2) · Label: 098006

✓ KNOWN MERSENNE

13

pattern bits

✓

p is prime

Hierarchical Decomposition:

098006 → p = 19937 (pattern: 0110111100001)

↳ Built from:

Filler zero

9 → p = 61 (pattern: 1101)

↳ Built from:

↺ Cycle: 9

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 21701 (base 2) · Label: 1507065

✓ KNOWN MERSENNE

13

pattern bits



p is prime

Hierarchical Decomposition:

1507065 → p = 21701 (pattern: 1010011000101)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

↺ Cycle: 1

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

p = 23209 (base 2) · Label: 95556

✓ KNOWN MERSENNE

13

pattern bits



p is prime

Hierarchical Decomposition:

95556 → p = 23209 (pattern: 1101010101001)

↳ Built from:

9 → p = 61 (pattern: 1101)

↳ Built from:

↺ Cycle: 9

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 44497 (base 2) · Label: 175906

✓ KNOWN MERSENNE

14

pattern bits

✓

p is prime

Hierarchical Decomposition:

175906 → p = 44497 (pattern: 10110111010001)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

↺ Cycle: 1

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

9 → p = 61 (pattern: 1101)

↳ Built from:

↺ Cycle: 9

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 86243 (base 2) · Label: 1500008007

✓ KNOWN MERSENNE

15

pattern bits

✓

p is prime

Hierarchical Decomposition:

1500008007 → p = 86243 (pattern: 101000011100011)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

↺ Cycle: 1

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

Filler zero

Filler zero

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

p = 110503 (base 3) • Label: 5785008

✓ KNOWN MERSENNE

15

pattern bits

✓

p is prime

Hierarchical Decomposition:

5785008 → p = 110503 (pattern: 010111110100111)

↳ Built from:

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

p = 132049 (base 2) • Label: 00000585005

✓ KNOWN MERSENNE

16

pattern bits

✓

p is prime

Hierarchical Decomposition:

00000585005 → p = 132049 (pattern: 0000001111010001)

↳ Built from:

Filler zero

Filler zero

Filler zero

Filler zero

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

8 → p = 31 (pattern: 111)

↳ Built from:

↻ Cycle: 8

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

Filler zero

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

p = 216091 (base 3) • Label: 507000077

✓ KNOWN MERSENNE

16

pattern bits

✓

p is prime

Hierarchical Decomposition:

507000077 → p = 216091 (pattern: 0100110000011011)

↳ Built from:

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

Filler zero

Filler zero

Filler zero

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

p = 756839 (base 2) · Label: 8007007008

✔ KNOWN MERSENNE

18

pattern bits

✔

p is prime

Hierarchical Decomposition:

$8007007008 \rightarrow p = 756839$ (pattern: 111000110001100111)

↳ Built from:

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

Filler zero

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

p = 859433 (base 3) • Label: 5000850556

✓ KNOWN MERSENNE

18

pattern bits

✓

p is prime

Hierarchical Decomposition:

5000850556 → p = 859433 (pattern: 010001110100101001)

↳ Built from:

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

Filler zero

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 1257787 (base 2) · Label: 707060087

✓ KNOWN MERSENNE

19

pattern bits

✓

p is prime

Hierarchical Decomposition:

707060087 → p = 1257787 (pattern: 0110011000100111011)

↳ Built from:

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

Filler zero

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

p = 1398269 (base 2) · Label: 155555885

✓ KNOWN MERSENNE

19

pattern bits

✓

p is prime

Hierarchical Decomposition:

155555885 → p = 1398269 (pattern: 1010101010111111101)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

↺ Cycle: 1

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

p = 2976221 (base 2) · Label: 975699

✓ KNOWN MERSENNE

20

pattern bits

✓

p is prime

Hierarchical Decomposition:

$975699 \rightarrow p = 2976221$ (pattern: 11010110100111011101)

↳ Built from:

$9 \rightarrow p = 61$ (pattern: 1101)

↳ Built from:

↺ Cycle: 9

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

$6 \rightarrow p = 17$ (pattern: 001)

↳ Built from:

↺ Cycle: 6

$9 \rightarrow p = 61$ (pattern: 1101)

↳ Built from:

↺ Cycle: 9

$9 \rightarrow p = 61$ (pattern: 1101)

↳ Built from:

↺ Cycle: 9

p = 3021377 (base 2) · Label: 80007560006

✓ KNOWN MERSENNE

20

pattern bits

✓

p is prime

Hierarchical Decomposition:

80007560006 → p = 3021377 (pattern: 11100001101001000001)

↳ Built from:

8 → p = 31 (pattern: 111)

↳ Built from:

↻ Cycle: 8

Filler zero

Filler zero

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

Filler zero

Filler zero

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

p = 6972593 (base 3) · Label: 550766706

✓ KNOWN MERSENNE

21

pattern bits



p is prime

Hierarchical Decomposition:

550766706 → p = 6972593 (pattern: 010100110010010110001)

↳ Built from:

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↻ Cycle: 5

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

p = 13466917 (base 3) · Label: 07508905055



KNOWN MERSENNE

22

pattern bits



p is prime

Hierarchical Decomposition:

07508905055 → p = 13466917 (pattern: 0011010111110100100101)

↳ Built from:

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

9 → p = 61 (pattern: 1101)

↳ Built from:

↺ Cycle: 9

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

p = 20996011 (base 2) · Label: 10000005088557



KNOWN MERSENNE

pattern bits



p is prime

Hierarchical Decomposition:

10000005088557 → p = 20996011 (pattern: 10000000101111110101011)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

↺ Cycle: 1

Filler zero

Filler zero

Filler zero

Filler zero

Filler zero

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

p = 24036583 (base 2) · Label: 1108706008008

 KNOWN MERSENNE

23

pattern bits



p is prime

Hierarchical Decomposition:

1108706008008 → p = 24036583 (pattern: 11011101100010011100111)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

↻ Cycle: 1

1 → p = 3 (pattern: 1)

↳ Built from:

↻ Cycle: 1

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↻ Cycle: 8

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

Filler zero

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↻ Cycle: 8

Filler zero

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↻ Cycle: 8

p = 25964951 (base 3) · Label: 0070007007608

✓ KNOWN MERSENNE

23

pattern bits

✓

p is prime

Hierarchical Decomposition:

0070007007608 → p = 25964951 (pattern: 00011000011000110010111)

↳ Built from:

Filler zero

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

Filler zero

Filler zero

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

Filler zero

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↻ Cycle: 7

6 → p = 17 (pattern: 001)

↳ Built from:

↻ Cycle: 6

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

🔄 Cycle: 8

p = 30402457 (base 3) · Label: 10588058076

✅ KNOWN MERSENNE

23

pattern bits

✅

p is prime

Hierarchical Decomposition:

10588058076 → p = 30402457 (pattern: 10011111110011110011001)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

🔄 Cycle: 1

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

🔄 Cycle: 5

8 → p = 31 (pattern: 111)

↳ Built from:

🔄 Cycle: 8

8 → p = 31 (pattern: 111)

↳ Built from:

🔄 Cycle: 8

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

🔄 Cycle: 5

8 → p = 31 (pattern: 111)

↳ Built from:

🔄 Cycle: 8

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 32582657 (base 3) · Label: 800505700000006

✓ KNOWN MERSENNE

23

pattern bits

✓

p is prime

Hierarchical Decomposition:

800505700000006 → p = 32582657 (pattern: 11100010010110000000001)

↳ Built from:

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

Filler zero

Filler zero

Filler zero

Filler zero

Filler zero

Filler zero

$6 \rightarrow p = 17$ (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 37156667 (base 2) · Label: 07758080087

✓ KNOWN MERSENNE

24

pattern bits

✓

p is prime

Hierarchical Decomposition:

$07758080087 \rightarrow p = 37156667$ (pattern: 001101101111011100111011)

↳ Built from:

Filler zero

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

p = 42643801 (base 2) · Label: 10055570005576

✓ KNOWN MERSENNE

24

pattern bits

✓

p is prime

Hierarchical Decomposition:

10055570005576 → p = 42643801 (pattern: 100010101011000101011001)

↳ Built from:

1 → p = 3 (pattern: 1)

↳ Built from:

↺ Cycle: 1

Filler zero

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

Filler zero

Filler zero

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

$6 \rightarrow p = 17$ (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 43112609 (base 2) · Label: 16000870055006

✓ KNOWN MERSENNE

24

pattern bits

✓

p is prime

Hierarchical Decomposition:

$16000870055006 \rightarrow p = 43112609$ (pattern: 100100011101100010100001)

↳ Built from:

$1 \rightarrow p = 3$ (pattern: 1)

↳ Built from:

↺ Cycle: 1

$6 \rightarrow p = 17$ (pattern: 001)

↳ Built from:

↺ Cycle: 6

Filler zero

Filler zero

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↻ Cycle: 8
7 → p = 19 (pattern: 011)
↳ Built from:

↻ Cycle: 7
Filler zero

Filler zero
5 → p = 13 (pattern: 01)
↳ Built from:

↻ Cycle: 5
5 → p = 13 (pattern: 01)
↳ Built from:

↻ Cycle: 5
Filler zero
Filler zero

6 → p = 17 (pattern: 001)
↳ Built from:
↻ Cycle: 6

p = 57885161 (base 3) · Label: 08009000058505

✔ KNOWN MERSENNE

24
pattern bits

✔

p is prime

Hierarchical Decomposition:

08009000058505 → p = 57885161 (pattern: 011100110100000111101001)
↳ Built from:

Filler zero
8 → p = 31 (pattern: 111)
↳ Built from:

↻ Cycle: 8
Filler zero
Filler zero

9 → p = 61 (pattern: 1101)
↳ Built from:

↺ Cycle: 9

Filler zero

Filler zero

Filler zero

Filler zero

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

$5 \rightarrow p = 13$ (pattern: 01)

↳ Built from:

↺ Cycle: 5

p = 74207281 (base 2) · Label: 077005500000706

✓ KNOWN MERSENNE

25

pattern bits

✓

p is prime

Hierarchical Decomposition:

$077005500000706 \rightarrow p = 74207281$ (pattern: 0011011000101000000110001)

↳ Built from:

Filler zero

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

Filler zero

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

Filler zero

Filler zero

Filler zero

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

p = 77232917 (base 2) • Label: 5009058700555

✓ KNOWN MERSENNE

25

pattern bits

✓

p is prime

Hierarchical Decomposition:

5009058700555 → p = 77232917 (pattern: 0100110100111101100010101)

↳ Built from:

5 → p = 13 (pattern: 01)

↳ Built from:

↺ Cycle: 5

Filler zero

Filler zero
 $9 \rightarrow p = 61$ (pattern: 1101)
↳ Built from:
↺ Cycle: 9

Filler zero
 $5 \rightarrow p = 13$ (pattern: 01)
↳ Built from:
↺ Cycle: 5
 $8 \rightarrow p = 31$ (pattern: 111)
↳ Built from:

↺ Cycle: 8
 $7 \rightarrow p = 19$ (pattern: 011)
↳ Built from:
↺ Cycle: 7

Filler zero
Filler zero
 $5 \rightarrow p = 13$ (pattern: 01)
↳ Built from:
↺ Cycle: 5
 $5 \rightarrow p = 13$ (pattern: 01)
↳ Built from:
↺ Cycle: 5
 $5 \rightarrow p = 13$ (pattern: 01)
↳ Built from:
↺ Cycle: 5

p = 82589933 (base 2) · Label: 0870000800080809

✓ KNOWN MERSENNE

25

pattern bits

✓

p is prime

Hierarchical Decomposition:

0870000800080809 → p = 82589933 (pattern: 0111011000011100011101101)

↳ Built from:

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

$7 \rightarrow p = 19$ (pattern: 011)

↳ Built from:

↺ Cycle: 7

Filler zero

Filler zero

Filler zero

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

Filler zero

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

$8 \rightarrow p = 31$ (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

$9 \rightarrow p = 61$ (pattern: 1101)

↳ Built from:

↺ Cycle: 9

p = 136279841 (base 2) · Label: 00007808086006

✓ KNOWN MERSENNE

26

pattern bits

✓

p is prime

Hierarchical Decomposition:

00007808086006 → p = 136279841 (pattern: 00000111110111011100100001)

↳ Built from:

Filler zero

Filler zero

Filler zero

Filler zero

7 → p = 19 (pattern: 011)

↳ Built from:

↺ Cycle: 7

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

Filler zero

8 → p = 31 (pattern: 111)

↳ Built from:

↺ Cycle: 8

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

Filler zero

Filler zero

6 → p = 17 (pattern: 001)

↳ Built from:

↺ Cycle: 6

 Known Mersenne Exponents (with hex labels)

1 (p=3)2 (p=5)3 (p=7)5 (p=13)6 (p=17)7 (p=19)8 (p=31)9 (p=61)a (p=89)b (p=107)c (p=127)0066 (p=521)578 (p=607)788 (p=1279)6077 (p=2203)696 (p=2281)6606 (p=3217)0669 (p=4253)650008
(p=4423)50876 (p=9689)77555 (p=9941)158075 (p=11213)098006 (p=19937)1507065 (p=21701)95556 (p=23209)175906 (p=44497)1500008007 (p=86243)5785008 (p=110503)00000585005
(p=132049)507000077 (p=216091)8007007008 (p=756839)5000850556 (p=859433)707060087 (p=1257787)155555885 (p=1398269)975699 (p=2976221)80007560006 (p=3021377)550766706

(p=6972593)07508905055 (p=13466917)10000005088557 (p=20996011)1108706008008 (p=24036583)0070007007608 (p=25964951)10588058076 (p=30402457)800505700000006 (p=32582657)07758080087
(p=37156667)10055570005576 (p=42643801)16000870055006 (p=43112609)08009000058505 (p=57885161)077005500000706 (p=74207281)5009058700555 (p=77232917)0870000800080809
(p=82589933)00007808086006 (p=136279841)